THE CONFORMAL OPERATOR IDENTITY BEHIND THE TEUKOLSKY EQUATION FOR EINSTEIN–HERMITIAN PERTURBATIONS

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Editorial note. The author proves this lemma in a named proof placed after the following Remark; the proof span is that named proof (CONTROL 8.8.157: a far-placed named proof is delivered as the single proof span).

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Editorial note. Referenced displays of the same paper are delivered as context spans so that the reader sees the equations the proof compares against (lane-s5 paper convention).

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Allowed standard prerequisites (editorial).

- GHP formalism and the weighted operators acting on scalar fields. Same paper, Section 2, in force in the statement. Fixes the notation in which the identity is written.
- The Teukolsky operator and the linearised Einstein equation. Same paper, the displayed equations preceding the lemma (statement span references them). Supplies the two sides of the identity.

$$E_{ab}(g) = R_{ab}(g) - \frac{S(g)}{4} g_{ab}, \tag{2.1}$$

$$\frac{\mathrm{d}}{\mathrm{d}s} g(s)_{ab}|_{s=0} = h_{ab}, \tag{2.2}$$

$$\vartheta\Psi_0 = \frac{\mathrm{d}}{\mathrm{d}s}\Psi_0[g(s)_{ab}]|_{s=0}, \tag{2.3}$$

2.2. **Spinors and complex structures.**

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2.3. **Conformally invariant GHP connections.**

$$f_a = -\frac{1}{2} J^c{}_b \nabla_c J^b{}_a, \tag{2.4}$$

$$P_a = \omega_a - \frac{1}{2}\mathrm{i} J^b{}_a f_b, \tag{2.5}$$

$$Q_{ab}{}^c = f_a \delta^c_b + f_b \delta^c_a - f^c g_{ab}.$$

$$o^A o^B \nabla_{AA'} o_B = 0 \qquad \Leftrightarrow \qquad \kappa = \sigma = 0, \tag{2.6}$$

$$\begin{aligned} \vartheta E_{ab} &= \vartheta R_{ab} - \frac{1}{4}\vartheta S g_{ab} - \frac{1}{4} S(g) h_{ab} \\ &= \frac{1}{2}\Delta h_{ab} - \frac{1}{2}\nabla_a \nabla_b \big(g^{cd} h_{cd}\big) + \frac{1}{2}\nabla^c \nabla_a h_{bc} + \frac{1}{2}\nabla^c \nabla_b h_{ac} \\ &\quad - \frac{1}{4} g_{ab}\big[\nabla^c \nabla^d h_{cd} + \Delta\big(g^{cd} h_{cd}\big) - h^{cd} R_{cd}\big] - \frac{1}{4} S(g) h_{ab}, \end{aligned} \tag{2.7}$$

$$\mathrm{L} = g^{ab} \mathcal{C}_a \mathcal{C}_b - 18\Psi_2 \tag{2.8}$$

Lemma 2.1. *Let (M, g_{ab}) be Einstein–Hermitian, with (possibly vanishing) cosmological constant λ. Let h_{ab} be an arbitrary metric perturbation* (2.2)*, ϑE_{ab} the linearized Einstein operator* (2.7)*, and $\vartheta\Psi_0$ the linearized Weyl scalar* (2.3)*. Furthermore, let f_a be the Lee form* (2.4)*, and Q^{abcd} the tensor field*

$$Q^{abcd} = o^A o^B o^C o^D \epsilon^{A'B'} \epsilon^{C'D'}. \tag{2.9}$$

Then

$$-\mathring{\Omega}^{-1} Q^{acbd} (\nabla_a - 4 f_a) \nabla_d \vartheta E_{bc} = \mathrm{L}\big[\mathring{\Omega}^{-1} \vartheta\Psi_0\big], \tag{2.10}$$

where L *is the operator* (2.8) *and $\mathring{\Omega}$ is an auxiliary constant conformal factor, that is a scalar field with weights $w = 1$, $p = 0$ and $\nabla_a \mathring{\Omega} = 0$.*

Proof of Lemma 2.1. The strategy is to consider identities for an arbitrary Riemannian manifold (M, g_{ab}), and then to linearize around an Einstein–Hermitian metric. Consider then an arbitrary (M, g_{ab}), with Levi-Civita connection ∇_a and unprimed Weyl curvature spinor Ψ_{ABCD}. Let $J^a{}_b$ be a (locally defined) compatible almost-complex structure, and let o^A be the associated spinor field as in Section 2.2. We can then define conformally and GHP weighted fields as in Section 2.3, together with the connection $\mathcal{C}_a$ on the corresponding bundles.

Let $\mathring{\Omega}$ be an arbitrary constant conformal factor, that is a scalar field with weights $w = 1$, $p = 0$ and $\nabla_a \mathring{\Omega} = 0$, and define

$$\varphi_{ABCD} := \mathring{\Omega}^{-1} \Psi_{ABCD}. \tag{2.11}$$

This object has weights $w = -1$, $p = 0$, and it is essentially the "gravitational spin 2 field" of Penrose and Rindler [3, equation (9.6.40)]. We have

$$\begin{aligned} \mathcal{C}_{AA'} \mathcal{C}^{A'E} \varphi_{BCDE} &= (\nabla_{AA'} - 4 f_{AA'}) \nabla^{A'E} \varphi_{BCDE} \\ &= \mathring{\Omega}^{-1} (\nabla_{AA'} - 4 f_{AA'}) \nabla^{A'E} \Psi_{BCDE} \\ &= \mathring{\Omega}^{-1} (\nabla_{AA'} - 4 f_{AA'}) \nabla^{B'}_{(B} \Phi_{CD)B'}{}^{A'}, \end{aligned}$$

where in the second line we used the definition (2.11) and the fact that $\mathring{\Omega}$ is constant, and in the third line we used the spinor form of the Bianchi identities [2,

equation (4.10.7)] adapted to Riemann signature (recall Section 2.2). Here $\Phi_{ABA'B'}$ is the trace-free Ricci spinor. Contracting with $o^A o^B o^C o^D$:

$$\begin{aligned} o^A o^B o^C o^D \mathcal{C}_{AA'} \mathcal{C}^{A'E} \varphi_{BCDE} &= \mathring{\Omega}^{-1} o^A o^B o^C o^D (\nabla_{AA'} - 4 f_{AA'}) \nabla_B^{B'} \Phi_{CDB'}{}^{A'} \\ &= \mathring{\Omega}^{-1} o^A o^C o^B o^D \epsilon^{A'C'} \epsilon^{B'D'} (\nabla_{AA'} - 4 f_{AA'}) \nabla_{DD'} \Phi_{BCB'C'} \\ &= -\frac{1}{2} \mathring{\Omega}^{-1} Q^{acbd} (\nabla_a - 4 f_a) \nabla_d E_{bc}, \end{aligned} \tag{2.12}$$

where in the last line we used the definition (2.9) and the identity $\Phi_{bc} = -\frac{1}{2} E_{bc}$ (see [2, equation (4.6.25)] and recall (2.1)).

It remains to find a convenient expression for the first line in (2.12). This can be done using [1, equations (3.6) and (3.15)]. We have

$$o^A o^B o^C o^D \mathcal{C}_{AA'} \mathcal{C}^{A'E} \varphi_{BCDE} = \frac{1}{2} \big(g^{ab} \mathcal{C}_a \mathcal{C}_b - 18 \Psi_2 \big) \varphi_0 + B, \tag{2.13}$$

where $\varphi_0 = o^A o^B o^C o^D \varphi_{ABCD} = \mathring{\Omega}^{-1} \Psi_0$ and B is a term which couples the GHP quantities κ, σ, Ψ_0, Ψ_1 quadratically. Equating the last line of (2.12) to the right-hand side of (2.13), and taking a linearization around a metric which satisfies $E_{ab}|_{s=0} = 0$, $\Psi_0|_{s=0} = \Psi_1|_{s=0} = \kappa|_{s=0} = \sigma|_{s=0} = 0$, that is, an Einstein–Hermitian metric, see (2.6), the result (2.10) follows. □

References

[1] Araneda B., Conformal invariance, complex structures and the Teukolsky connection, *Classical Quantum Gravity* **35** (2018), 175001, 23 pages, arXiv:1805.11600.

[2] Penrose R., Rindler W., Spinors and space-time. Vol. 1. Two-spinor calculus and relativistic fields, *Cambridge Monogr. Math. Phys.*, Cambridge University Press, Cambridge, 1984.

[3] Penrose R., Rindler W., Spinors and space-time. Vol. 2. Spinor and twistor methods in space-time geometry, *Cambridge Monogr. Math. Phys.*, Cambridge University Press, Cambridge, 1986.